



VISAYAS
STATE UNIVERSITY



Lesson 2.1 - Linear Equations

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Objectives

At the end of the lesson, the students will be able to:

- Define equations.
- Define linear equations.
- Solve linear equations in one variable.
- Solve word problems involving linear equations in one variable.

Equations

An **equation** is a statement that two mathematical expressions are equal, e.g., $3 + 5 = 8$ is an equation.

Most equations that we study in algebra contain **variables**, which are symbols (usually letters) that stand for numbers. In the equation,

$$4x + 7 = 19$$

the letter x is the variable. We think of x as the “unknown” in the equation, and our goal is to find the value of x that makes the equation true. The values of the unknown that make the equation true are called the **solutions** or **roots** of the equation, and the process of finding the solutions is called **solving the equation**.

Two equations with exactly the same solutions are called **equivalent equations**. To solve an equation, we try to find a simpler, equivalent equation in which the variable stands alone on one side of the “equal” sign.

PROPERTIES OF EQUALITY

Property

1. $A = B \Leftrightarrow A + C = B + C$

2. $A = B \Leftrightarrow CA = CB \quad (C \neq 0)$

Description

Adding the same quantity to both sides of an equation gives an equivalent equation.

Multiplying both sides of an equation by the same nonzero quantity gives an equivalent equation.

These properties require that you perform the same operation on both sides of an equation when solving it. Thus if we say “add 7” when solving an equation, that is just a short way of saying “add 7 to each side of the equation.”

Linear Equations

Definition

The simplest type of equation is a **linear equation**, or first-degree equation, which is an equation in which each term is either a constant or a nonzero multiple of the variable.

Definition

A **linear equation in one variable** is an equation equivalent to the form

$$ax + b = 0$$

where a and b are real numbers, $a \neq 0$, and x is a variable.

Linear equation	Nonlinear equation
$4x - 5 = 3$	$x^2 + 2x = 8$
$2x = \frac{1}{2}x - 7$	$\sqrt{x} - 6x = 0$
$x - 6 = \frac{x}{3}$	$\frac{3}{x} - 2x = 1$

STEPS TO FOLLOW IN SOLVING A LINEAR EQUATION IN ONE UNKNOWN:

- ➊ If the equation is fractional, clear the equation of fractions by multiplying both sides of the equation by the Least Common Multiple of all the denominators (LCM).
- ➋ Simplify both sides of the equation by carrying out indicated operations of multiplication if there are any and by collecting like terms.
- ➌ Put all the terms containing the unknown to one side of the equation and all the other terms on the other side.
- ➍ Divide both sides of the equations by the coefficients of the unknown.
- ➎ Check the solution by substituting the value obtained in the original equation. Reject all extraneous roots.

Linear Equations

Example

Tell whether the equation is linear or not if linear solve for the roots

① $\frac{3x - 5}{5} + \frac{x + 3}{4} = 6$

② $\frac{0.4x}{9} + 0.2x = \frac{2.2}{3}$

③ $0.05(x + 2) = 0.03(2x + 4)$

④ $\frac{2}{x + 5} + \frac{1}{x - 5} = \frac{16}{x^2 - 25}$

⑤ $\frac{3}{y - 2} + \frac{2y}{4 - y^2} = \frac{5}{y + 2}$

FOR EQUATIONS INVOLVING RADICALS

- 1 Put one of the radicals (if there are more than one) on one side of the equation and all the other terms on the other side by transposition.
- 2 Square both sides and simplify.
- 3 Repeat steps 1 and 2 (if necessary) until the resulting equation has been cleared of radicals.
- 4 Solve the resulting equations.
- 5 Check by substituting the value obtained to the original equation and discard extraneous roots.

Example

Tell whether the equation is linear or not if linear solve for the roots

❶ $\sqrt[3]{x+5} = 2$

❷ $\sqrt{3y+1} = \sqrt{2y+6}$

❸ $2\sqrt{t-1} = \sqrt{3t-1}$

❹ $\sqrt{2x-3} = 1 + \sqrt{x+2}$

❺ $\sqrt{x+5} + 2 = \sqrt{2x+5}$

Linear Equations

Example (Verbal problems)

- 1 Narda and Rosie went to their backyard to pick flowers together they picked 60 flowers. Rosie picked 12 more than half the number of flowers picked by Narda. How many flowers did each girl picked?
- 2 A farmer can plow a field in 2 days using a tractor while his brother can make it in 8 days using a plow. If they work together, how many days can they finish the job?
- 3 Two cars started from the same place and 90 in opposite direction. One travel 12 km faster than the other. Find the spend of each if they are 324 km apart after 3 hours.
- 4 A car rental company charges Php1,500 a day and Php7.5 a mile for renting a car. Helen rents a car for two days, and her bill comes to Php5,400. How many miles did she drive?
- 5 Mary inherits Php5,000,000 and invests it in two certificates of deposit. One certificate pays 6% and the other pays 4.5% simple interest annually. If Mary's total interest is Php251,250 per year, how much money is invested at each rate?